

Industrial Training Logbook Year: 2024

Student's Name: Nada Ashraf Moussa Kamel Gomaa

Student's ID: 120210358

Declaration

Before submitting this document for assessment, please complete the following declaration:

This is to certify that the contents of the Industrial Training Logbook are a true and accurate reflection of the work done by the student in the training company.

Student's name: Nada Ashraf Moussa Kamel Gomaa

Student's signature: Nada Ashraf Moussa Kamel Gomaa

Supervisor's name: Egypt-Digital-Pioneers-Initiative

Supervisor's signature: DEPI- Dr. Eman Raslan

Company's name: CLS- Learning Solutions

Student's Details

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Major: Computer Science and Engineering

Minor: ECCE

Placement Details

Name of Company: Egypt Digital Pioneers Initiative (CLS)

Company Address: 26 Sharm El-Dokki Main Street, Next to Misr Gas Station in front of Dokki Metro Station, Dokki, Giza Governorate

Industrial Supervisor: CLS learning solutions

Company Contact Tel No.: +20 11 53615270 **Fax No.:** +20 11 53615270

Training Period: 7 Months (April to October 2024)

Allowance: –

(Indicate the amount of allowance paid per month)

Student's Logbook

Training intensity assumption used for planning: 3–4 meetings per week (3–4 hours each) + continuous labs and self-study.

Student's Logbook**Date:** 1 April – 14 April 2024**Weeks 1–2**

Activities:	• Orientation to the training structure and deliverables (logbook discipline, weekly reporting, and lab submission workflow). • Orientation to Microsoft SQL and relational database fundamentals. • Learning SQL essentials: DDL (CREATE/ALTER/DROP) and DML (SELECT/INSERT/UPDATE/DELETE). • Hands-on: database setup and environment configuration for practice. • Hands-on: writing simple queries with filters, sorting, grouping, and joins. • Short exercises in query debugging, fixing syntax/logic errors, and validating outputs using sample data.
Summary:	Initial phase focused on SQL foundations and establishing a correct working environment, followed by intensive practice on essential query patterns required for data engineering and analytics workflows.
Key Topics:	• Relational databases and SQL Server environment setup • DDL vs DML and correct usage • Core SELECT queries, WHERE, ORDER BY, GROUP BY • JOIN types (INNER/LEFT/RIGHT) and practical use cases • Query troubleshooting and result validation
Achievement of the Day: Developed foundational skills in computer usage, problem-solving, teamwork, and communication	
Functional Skills:	• Computer Skills • Problem-solving Skills
Soft Skills:	• Communication Skills • Teamwork Skills

Student's Logbook**Date:** 15 April – 28 April 2024**Weeks 3–4**

Activities:

- Advanced SQL practice: multi-table joins, subqueries, correlated queries, and common table expressions (CTEs).
- Data quality logic in SQL: NULL handling, constraints awareness, and consistency checks.
- Performance basics: understanding indexes at a conceptual level and why queries slow down.
- Applied tasks: create tables + insert sample data + write analytical queries that simulate business reporting.
- Documentation habit: writing query comments, assumptions, and expected outputs for reproducibility.

Summary: Deepened SQL capability from basic querying to more complex analytical and engineering-oriented patterns, emphasizing correctness, validation, and early performance awareness.

Key Topics:

- CTEs, subqueries, correlated queries
- Aggregation with HAVING and multi-step analytics
- Data validation logic (NULL, missing values, duplicates conceptually)
- Basic query performance considerations and indexing concepts
- Reproducible SQL documentation practices

Achievement of the Day: Advanced SQL proficiency including complex queries, CTEs, and performance optimization techniques

Functional Skills:

- Organizing/Analyzing Data
- Decision-making Ability

Soft Skills:

- Critical Thinking Skills
- Teamwork Skills

Student's Logbook**Date:** 29 April – 12 May 2024**Weeks 5–6**

Activities:

- Implementing SQL Data Warehouse concepts: OLTP vs OLAP and why analytics requires different schema design.
- Dimensional modeling practice: fact tables, dimension tables, and star schema vs snowflake schema.
- Normalization vs denormalization trade-offs and when each is justified.
- Hands-on: sketching a small warehouse schema for analytics reporting scenarios.
- Designed example KPIs and mapped them to facts/measures and dimensions.

Summary: Shifted from operational SQL usage to analytical system design: learned how warehouse schemas support reporting performance, consistent metrics, and scalable analytics.

Key Topics:

- Data warehouse fundamentals (OLTP vs OLAP)
- Fact vs Dimension tables and grain definition
- Star vs snowflake schema
- Normalization vs denormalization trade-offs
- KPI mapping to schema (measures/dimensions)

Achievement of the Day: Understanding of data warehouse architecture and dimensional modeling for analytics

Functional Skills:

- Organizing/Analyzing Data
- Decision-making Ability

Soft Skills:

- Teamwork Skills
- Critical Thinking Skills

Student's Logbook**Date:** 13 May – 26 May 2024**Weeks 7–8**

Activities:	• ETL (Extract, Transform, Load) process concepts and why ETL is essential for warehouse reliability. • Data pipeline steps: extracting from sources, cleaning, transforming, and loading into target schema. • Data integrity and quality checks: consistency rules, duplicate handling, missing data strategies. • Practical mini-pipeline: simulate ETL with SQL scripts and structured transformation steps. • Team-based review sessions: comparing schema designs, discussing transformation rules and validation logic.
Summary:	Built a structured understanding of ETL pipelines and quality controls required to maintain trustworthy analytics datasets in a warehouse environment.
Key Topics:	• ETL workflow and stages • Data quality rules and validation checks • Transformations aligned to business definitions • Loading strategies and schema alignment • Documentation of assumptions and pipeline steps
Achievement of the Day: Practical ETL pipeline development and data quality management skills	
Functional Skills:	• Organizing/Analyzing Data • Problem-solving Skills
Soft Skills:	• Communication Skills • Teamwork Skills

Student's Logbook	
Date:	27 May – 9 June 2024
Weeks 9–10	
Activities:	• Python programming fundamentals for data workflows: syntax, variables, loops, functions. • Data structures: lists, tuples, dictionaries, sets; selecting structures based on task needs. • Error handling with try/except; writing safer scripts. • File handling and directory operations: reading/writing files, working with paths. • Initial work with NumPy arrays and basic Pandas operations for data manipulation.
Summary:	Established Python fundamentals required for automation, data preprocessing, and future ML experimentation. Emphasis was placed on correctness, readable code, and controlled error handling.
Key Topics:	• Python basics (functions, loops, conditions) • Data structures and practical usage • Error handling and debugging workflow • File handling and scripting habits • NumPy and Pandas basics
Achievement of the Day: Foundation in Python programming for data workflows and automation	
Functional Skills:	• Computer Skills • Quantitative & Analytical Skills
Soft Skills:	• Communication Skills • Interpersonal Skills

Student's Logbook**Date:** 10 June – 23 June 2024**Weeks 11–12**

Activities:	• Data preprocessing with Python: handling missing values, duplicates conceptually, encoding categorical data (concept-level). • Exploratory Data Analysis basics: summary stats, simple plots, and interpretation. • Intro to ML workflow idea: data → split → train → evaluate (concept-level). • Strengthened notebook discipline: clean sections, comments, and reproducible runs. • Group sessions: presenting solutions, peer review of code clarity and correctness.
Summary:	Expanded Python skills toward data preparation and analysis patterns that directly support ML and data engineering deliverables.
Key Topics:	• Data preprocessing patterns in Python • Basic EDA workflow and interpretation • Reproducible notebook structure and documentation • Clean coding practices and peer review • Intro ML pipeline framing (conceptual)
Achievement of the Day: Data preprocessing and exploratory analysis skills for machine learning readiness	
Functional Skills:	• Organizing/Analyzing Data • Problem-solving Skills
Soft Skills:	• Critical Thinking Skills • Teamwork Skills

Student's Logbook**Date:** 24 June – 7 July 2024**Weeks 13–14**

Activities:	● Introduction to Agile development and Scrum roles/events/artifacts. ● Participation in sprint planning and daily stand-up simulations. ● Understanding tickets/issue tracking systems: breaking tasks, estimating, defining acceptance criteria. ● Documentation of sprint tasks and aligning technical deliverables with sprint goals. ● Collaboration practice: team communication rules, meeting notes, and action items.
Summary:	Learned Agile/Scrum as an operational framework for managing technical tasks, coordinating teams, and ensuring traceable progress across complex training modules and project milestones.
Key Topics:	● Agile principles and Scrum framework ● Sprint planning, daily standups, task breakdown ● Ticketing workflow and acceptance criteria ● Documentation habits for engineering projects ● Collaboration and coordination routines
Achievement of the Day: Practical ETL pipeline development and data quality management skills	
Functional Skills:	● Organizing/Analyzing Data ● Problem-solving Skills
Soft Skills:	● Communication Skills ● Teamwork Skills

Student's Logbook**Date:** 8 July – 21 July 2024**Weeks 15–16**

Activities:

- Azure Data Fundamentals: core cloud concepts and why cloud matters for data/ML systems.
- Understanding cloud storage, compute, and security concepts; shared responsibility concept-level.
- Hands-on with Azure Data Studio: connecting, exploring data, and running queries.
- Relational vs non-relational databases: use-cases and tradeoffs.
- Security and compliance awareness for data systems (concept-level).

Summary: Developed baseline cloud literacy focused on data platforms, including tool usage (Azure Data Studio) and fundamental design/security considerations.

Key Topics:

- Cloud data concepts and service categories
- Azure Data Studio workflow
- Relational vs non-relational databases
- Security/compliance awareness (fundamentals)
- Cloud value for business and scalability

Achievement of the Day: Cloud data platform fundamentals and Azure Data Studio proficiency

Functional Skills:

- Computer Skills
- Decision-making Ability

Soft Skills:

- Communication Skills
- Interpersonal Skills

Student's Logbook**Date:** 22 July – 4 August 2024**Weeks 17–18**

Activities:

- Advanced Azure discussion: designing secure and scalable data solutions (concept-level).
- Data ingestion and pipeline thinking: batch vs streaming (concept-level).
- Monitoring and optimization mindset for cloud workflows (concept-level).
- Team-based case studies: mapping business requirements to data services and pipeline components.
- Consolidation quiz sessions and practical review.

Summary: Moved from fundamental cloud concepts to solution-thinking: how to select services and plan pipelines while considering scalability, monitoring, and security.

Key Topics:

- Designing data solutions in Azure (concepts)
- Batch vs streaming ingestion (overview)
- Monitoring and optimization mindset
- Security and scalability considerations
- Translating requirements to architecture

Achievement of the Day: Advanced cloud solution design thinking and architecture planning

Functional Skills:

- Organizing/Analyzing Data
- Decision-making Ability

Soft Skills:

- Leadership Skills
- Communication Skills

Student's Logbook**Date:** 5 August – 18 August 2024**Weeks 19–20****Activities:**

- Course 7: NLP scope and pipeline: text acquisition, cleaning, normalization, tokenization, modeling, evaluation.
- NLU vs NLG: understanding vs generation tasks and examples.
- NLP applications covered: toxicity classification, topic classification, grammatical error correction, information retrieval, question answering.
- Key NLP challenges: ambiguity (lexical/syntactic/semantic), context dependence, noisy web data, multilinguality/low-resource languages, implicit meaning (sarcasm/cultural context).
- Preprocessing toolkit concepts: lowercase tradeoffs, regex concept practice, tokenization types (character/word/sentence/subword).
- Tool comparison: NLTK vs spaCy (use-cases, tokenization support, pipeline differences).

Summary:

Built the foundations of NLP workflow and challenges, linking preprocessing decisions to downstream model performance and evaluation reliability.

Key Topics:

- NLP pipeline and task taxonomy (NLU/NLG)
- Text cleaning and normalization decisions
- Tokenization types and trade-offs
- NLTK vs spaCy overview
- Real-world NLP application mapping

Achievement of the Day: NLP pipeline understanding and text preprocessing proficiency**Functional Skills:**

- Problem-solving Skills
- Quantitative & Analytical Skills

Soft Skills:

- Critical Thinking Skills
- Teamwork Skills

Student's Logbook**Date:** 19 August – 1 September 2024**Weeks 21–22**

Activities:	• Feature engineering for text: one-hot encoding (limitations), Bag of Words, TF-IDF, and N-grams. • Understanding sparsity/dimensionality problems and why vocabulary control matters. • Traditional ML modeling for NLP: logistic regression, naive Bayes, decision trees (concept + workflow). • Practice workflow: vectorizer → model → evaluation (concept-level). • Lab-style discussion based on supervised classification pipelines (e.g., fake news-style classification as a practice theme).
Summary:	Advanced from preprocessing to classical NLP representations and traditional ML models, emphasizing interpretability, pipeline structure, and evaluation discipline.
Key Topics:	• One-hot, BoW, TF-IDF, N-grams • Sparsity and dimensionality concerns • Classical supervised NLP models • Vectorizer-to-classifier pipelines • Evaluation framing and error analysis mindset
Achievement of the Day: Classical NLP techniques and feature engineering for text data	
Functional Skills:	• Organizing/Analyzing Data • Decision-making Ability
Soft Skills:	• Teamwork Skills • Critical Thinking Skills

Student's Logbook**Date:** 2 September – 15 September 2024**Weeks 23–24**

Activities:	• Deep learning NLP evolution: embeddings and sequence modeling. • Embeddings: Word2Vec (CBOW/Skip-gram), GloVe, FastText; contextual embeddings concept: ELMo/BERT/GPT/T5. • RNN fundamentals and gradient issues; motivation for LSTM and GRU. • Seq2Seq concept: encoder/decoder, bottleneck problem, and attention mechanism concept. • Evaluation metrics overview: BLEU (translation), ROUGE (summarization), WER (speech). • Transformers overview and why attention is important for long-range dependencies.
Summary:	Covered deep-learning NLP foundations and the motivation for transformer architectures, linking model families to tasks and evaluation metrics.
Key Topics:	• Static vs contextual embeddings • RNN/LSTM/GRU concept and limitations • Seq2Seq and attention concept • BLEU/ROUGE/WER evaluation overview • Transformer motivation and attention role
Achievement of the Day: Deep learning NLP foundations and transformer architecture understanding	
Functional Skills:	• Quantitative & Analytical Skills • Problem-solving Skills
Soft Skills:	• Leadership Skills • Communication Skills

Student's Logbook	
Date:	16 September – 29 September 2024
Weeks 25–26	
Activities:	• Course 8: Hugging Face overview and ecosystem components: Hub, Transformers, Datasets, Tokenizers, Spaces, Accelerate. • Hub usage concepts: model hub, dataset hub, model/dataset cards, hosted widgets for quick testing. • Transformers library: pre-trained model families and tasks (classification, generation, QA, translation, NER, summarization). • Pipelines abstraction: tokenization + inference + post-processing as an end-to-end workflow. • Model architecture families mapping: – Encoder: BERT/DistilBERT/RoBERTa for classification/NER/extractive QA. – Decoder: GPT-style for text generation. – Encoder-decoder: T5/BART/mBART/Marian for translation/summarization/generative QA. • Discussed transformer challenges: language dominance (English), data availability, long-document cost, opacity, bias in internet-trained models.
Summary:	Established working knowledge of Hugging Face tooling and how its libraries and Hub accelerate experimentation, sharing, and deployment for transformer-based workflows.
Key Topics:	• Hugging Face Hub (models/datasets/spaces) and cards • Transformers library and pipelines • Datasets and Tokenizers libraries • Model family selection by task • Known transformer limitations: bias/opacity/long context
Achievement of the Day: Hugging Face ecosystem proficiency and transformer model selection skills	
Functional Skills:	• Computer Skills • Organizing/Analyzing Data
Soft Skills:	• Communication Skills • Teamwork Skills

Student's Logbook**Date:** 30 September – 13 October 2024**Weeks 27–28**

Activities:	● MLOps definition: practices to manage ML development, deployment, maintenance with automation across the lifecycle. ● Why MLOps: scalability, reproducibility, fast time-to-market, and monitoring. ● Core data governance concepts: data lineage, data provenance, metadata. ● CI/CD in MLOps: automated testing of code and models, version control for code/data/models, automated data validation. ● Continuous Training (CT): retraining when data profile changes and drift affects performance. ● MLflow overview: Tracking, Projects, Models, Model Registry. ● MLflow Tracking: logging parameters, metrics, artifacts and using UI for comparison. ● MLflow Projects: packaging code + environment for reproducibility (Conda/Docker concept). ● MLflow Models: packaging and deployment options (local, REST, cloud/container concept). ● MLflow Model Registry: versioning + staging + production + archived lifecycle control.
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Summary:	Built MLOps literacy centered on MLflow as a lifecycle management tool, connecting governance and CI/CD/CT concepts to reliable ML delivery.
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Key Topics:	● MLOps lifecycle and automation rationale ● Data lineage/provenance/metadata ● CI/CD adaptations for ML + Continuous Training ● MLflow Tracking/Projects/Models/Registry ● Reproducibility and versioning practices
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Achievement of the Day: MLOps practices and MLflow lifecycle management understanding

Functional Skills:	● Quantitative & Analytical Skills ● Problem-solving Skills
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Soft Skills:	● Critical Thinking Skills ● Teamwork Skills
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Student's Logbook

Date: 14 October – 27 October 2024

Weeks 29–30

Activities:

- Course 9 (Prompt Engineering): prompt structure, constraints, format control, examples (zero-shot vs few-shot), iterative refinement and evaluation prompts.
- Applied prompt engineering for practical work: debugging assistance, refactoring notebooks, generating checklists and edge cases, improving documentation wording and structure.
- Final applied project: **cheXray_Classifier** (Pneumonia vs Normal) using Kaggle dataset:
 - Dataset: 5,863 pediatric chest X-rays; folders train/test/val with NORMAL/PNEUMONIA.
 - Context and reference paper link included in resources.
- Colab pipeline implementation (end-to-end):
 - Kaggle API setup: upload kaggle.json, permissions, download + unzip dataset.
 - Verified counts per split and label folder; validated directory structure.
 - Built file path lists using os.walk; shuffled; confirmed lengths.
 - Preprocessing and augmentation with ImageDataGenerator: rescale, rotation, shifts, shear, zoom, horizontal flip, fill mode.
 - Generators via flow_from_directory with target_size=(224,224), batch_size=32, class_mode=binary.
 - Visual sanity checks: plotting sample batches from train/val/test generators.
- Modeling work (baseline + transfer learning + fine-tuning):
 - Baseline custom CNN (Functional API): Conv2D blocks (16/32/64), BatchNorm, ReLU, MaxPool, Dropout, Flatten, Dense, Sigmoid.
 - Callbacks: ModelCheckpoint (best weights), EarlyStopping, ReduceLROnPlateau.
 - Transfer learning: ResNet152V2 (ImageNet, include_top=False), frozen base, custom head (GAP, Dense, BatchNorm, Dropout, Sigmoid).
 - Fine-tuning: unfreeze base model with controlled layer freezing (e.g., freeze all but last 30 layers), careful LR settings, monitoring overfitting.
- Deployment workflow (Hugging Face + Gradio):
 - Hugging Face login (CLI/notebook), repository use, versioning with add/commit/push concept.
 - Model saving (.h5/.keras) and packaging for inference.
 - Gradio demo: image upload → resize to 224x224 → normalize → predict → label (Pneumonia/Normal) with threshold 0.5.
 - Validated end-to-end inference usability.
- Final presentation preparation: problem statement, dataset context, preprocessing, models, evaluation approach, deployment screenshots/steps, and lessons learned.

Summary:

Completed an end-to-end ML project under an intensive schedule: dataset ingestion, preprocessing, baseline CNN, transfer learning + fine-tuning, and deployment using Hugging Face/Gradio, supported by MLOps and prompt engineering practices.

Helpful References (Links)

- Kaggle dataset (Chest X-Ray Pneumonia): <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>
- Dataset clinical context / paper page (Cell): [http://www.cell.com/cell/fulltext/S0092-8674\(18\)30154-5](http://www.cell.com/cell/fulltext/S0092-8674(18)30154-5)
- TensorFlow ImageDataGenerator docs: https://www.tensorflow.org/api_docs/python/tf/keras/preprocessing/image/ImageDataGenerator
- TensorFlow Transfer Learning guide: https://www.tensorflow.org/tutorials/images/transfer_learning
- Keras Callbacks (EarlyStopping / ReduceLROnPlateau / ModelCheckpoint): <https://keras.io/api/callbacks/>
- Hugging Face main site: <https://huggingface.co/>
- Hugging Face Hub docs: <https://huggingface.co/docs/hub/index>
- Hugging Face Transformers docs: <https://huggingface.co/docs/transformers/index>
- Hugging Face Spaces docs: <https://huggingface.co/docs/hub/spaces>
- Gradio documentation: <https://www.gradio.app/docs>
- MLflow official documentation: <https://mlflow.org/docs/latest/index.html>